



Storm Sequence Generator — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 29-March-2026

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1 Test Results — Storm Sequence Generator (MKM-SS-001)

Tests run on **29 March 2026 at 10:29:14**.

Table 1: Test Results Summary — Storm Sequence Generator (MKM-SS-001)

Total Tests	17
Passed	17
Failed	0
Skipped	0
Duration	0.00s

Table 2: Detailed Test Results — Storm Sequence Generator

Test Description	Acceptance Criteria	Result	Time
Duration must not exceed MAX_PRECIP_END_HOUR (156).	<code>result.storms[0].duration_hours <= MAX_PRECIP_END_HOUR</code>	Pass	0.000s
Isolated has one storm	<code>isinstance(result, StormSequence);</code> <code>result.num_storms == 1 (+2 more)</code>	Pass	0.000s
Aggregate stats	<code>result.max_intensity_factor >=</code> <code>result.avg_intensity_factor;</code> <code>result.total_precipitation_mm > 0</code> (+1 more)	Pass	0.000s
Catchment id in storms	<code>result.storms[0].catchment_id ==</code> "rhine"	Pass	0.000s
Cluster sequence	<code>result.sequence_type in</code> (SequenceType.CLUSTER.value, Sequ...	Pass	0.000s
force_sequence_type='doublet' → always doublet.	<code>result.sequence_type ==</code> SequenceType.DOUBLET.value; <code>result.num_storms == 2</code>	Pass	0.000s
force_sequence_type='isolated' → always isolated.	<code>result.sequence_type ==</code> SequenceType.ISOLATED.value; <code>result.num_storms == 1</code>	Pass	0.000s
Minimal intensity (5% sequence probability) mostly produces isolated.	<code>isolated >= 10</code>	Pass	0.001s
Generate multiple sequences without error.	<code>isinstance(result, StormSequence);</code> <code>result.num_storms >= 1</code>	Pass	0.000s
Persistent sequence	<code>result.sequence_type in</code> (SequenceType.PERSISTENT.value, S...	Pass	0.000s
Returns storm sequence	<code>isinstance(result, StormSequence)</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
All sequences must have to- tal_duration != 168h.	result.total_duration_hours <= 168.0	Pass	0.000s
Storms have correct fields	storm.duration_hours > 0; storm.intensity_factor > 0 (+2 more)	Pass	0.000s
Custom catchment	gen.catchment_id == "rhine"; gen.base_precipitation == CATCHMENT_BASE_PRECIP["rhine"]	Pass	0.000s
Default catchment	gen.catchment_id == "thames"; gen.base_precipitation == CATCHMENT_BASE_PRECIP["thames"]	Pass	0.000s
Same seed → same sequence.	seq1.num_storms == seq2.num_storms; seq1.sequence_type == seq2.sequence_type	Pass	0.000s
Unknown catchment defaults to 35	gen.base_precipitation == 35.0	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
12-Mar-2026	1.0	Initial test suite (116 tests)	D. Kelly
27-Mar-2026	1.1	64 test(s) removed (total: 52)	D. Kelly
29-Mar-2026	1.2	35 test(s) removed (total: 17)	D. Kelly